RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision (Delta)
Topic(s): (C) Statistics
(Solutions)

Source of Question: JPJC Prelim 9758/2025/P2/Q8

1 A company makes portable speakers. Some speakers turn out to be faulty. Each carton has 24 portable speakers.

(i) State, in context, two assumptions needed for the number of faulty portable speakers in a randomly chosen carton to be well modelled by a binomial distribution. [2]

It is given that 2% of the portable speakers produced are faulty.

(ii) Find the probability that a randomly chosen carton contains at most one faulty portable speaker. [1]

(iii) A carton with more than one faulty portable speaker is considered as 'substandard'. Find the probability that, out of 50 cartons, there are no 'substandard' cartons. [2]

The company also makes flash drives packed in batches of one hundred. The probability that a flash drive being faulty is p, where $0 < p < 1$. For quality control purposes, a random sample of 3 flash drives is tested from each batch.

- If there are 2 or 3 faulty flash drives, the batch is rejected.
- If there is no faulty flash drive, the batch is accepted.
- If there is one faulty flash drive in the sample, another random sample of 3 flash drives is tested and the batch is accepted if there are no faulty flash drive in the second sample. Otherwise the batch is rejected.

(iv) Find the range of p such that we accept at least 95% of the batches based on this quality control method. [3]

Solution:

(i)	(1) The **event** that a portable speaker is faulty **is independent of** the **event** that another portable speaker is faulty. (2) The **probability** of a portable speaker being faulty **remains constant** for all portable speakers.
(ii)	Let X denote the **number** portable speakers, out of 24, that are faulty. $X \sim \mathrm{B}(24, 0.02)$ $\mathrm{P}(X \le 1) \approx 0.917387 = 0.917$ (3 s.f.)
(iii)	**<u>Method 1:</u>** Let Y denote the **number** of cartons, out of 50, that are substandard. P(carton is substandard) = $\mathrm{P}(X > 1) \approx 1 - 0.917387 = 0.082613$ $Y \sim \mathrm{B}(50, 0.082613)$ $\mathrm{P}(Y = 0) \approx 0.013416 = 0.0134$ (3 s.f.) **<u>Method 2:</u>** Required Probability $= \left[\mathrm{P}(X \le 1)\right]^{50} \approx \left[0.917387\right]^{50} = 0.0134$ (3 s.f)

(iv)

Let F denote the **number** of flash drives, out of 3, that are faulty.

$F \sim \mathrm{B}(3, p)$

P(batch is accepted)

$= \mathrm{P}(F=0) + \mathrm{P}(F=1) \times \mathrm{P}(F=0)$

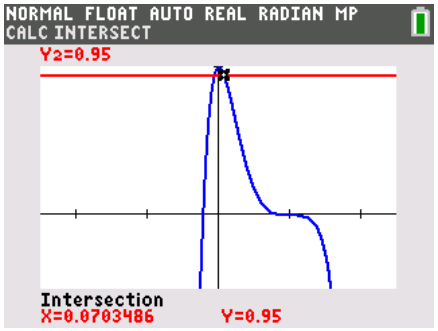
$= (1-p)^3 + {}^3C_1 p^1 (1-p)^2 \times (1-p)^3$

$= (1-p)^3 \left[1 + 3p(1-p)^2\right]$

If at least 95% of the batches are accepted,

$(1-p)^3 \left[1 + 3p(1-p)^2\right] \geq 0.95$

From the GC, $0 < p \leq 0.0703$.



Source of Question: CJC Prelim 9758/2022/02/Q7

2 The probability distribution for the random variable X is shown in the table.

x	-1	0	1	2
$\mathrm{P}(X=x)$	p	$\frac{1}{5}$	q	$\frac{1}{5}$

Another random variable Y is defined by $Y = 2X_1 + X_2$, where X_1 and X_2 are independent observations of X.

(i) Given that $\mathrm{P}(Y=4) = \frac{3}{25}$, find the values of p and q. [3]

(ii) Find $\mathrm{E}(X)$ and $\mathrm{Var}(X)$. [2]

(iii) Find $\mathrm{Var}(Y)$. [2]

Solution:

(i)

$p + q + \frac{2}{5} = 1 \Rightarrow p + q = \frac{3}{5}$ ---(1)

$\mathrm{P}(Y=4) = \frac{3}{25}$

$\mathrm{P}(X_1 = 1, X_2 = 2) + \mathrm{P}(X_1 = 2, X_2 = 0) = \frac{3}{25}$

$q \times \frac{1}{5} + \frac{1}{5} \times \frac{1}{5} = \frac{3}{25}$

$\frac{1}{5} q = \frac{2}{25}$

$q = \frac{2}{5}$

Substitute $q = \frac{2}{5}$ into equation (1), $p + \frac{2}{5} = \frac{3}{5} \Rightarrow p = \frac{1}{5}$

(ii)

$$\mathrm{E}(X) = (-1)\frac{1}{5} + (1)\frac{2}{5} + (2)\frac{1}{5} = \frac{3}{5}$$

$$\mathrm{E}(X^2) = (-1)^2\left(\frac{1}{5}\right) + (1)^2\left(\frac{2}{5}\right) + (2)^2\left(\frac{1}{5}\right) = \frac{7}{5}$$

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 = \frac{7}{5} - \left(\frac{3}{5}\right)^2 = \frac{35}{25} - \frac{9}{25} = \frac{26}{25}$$

OR Using GC:



From GC, $\mathrm{E}(X) = 0.6$ and $\mathrm{E}(X^2) = 1.4$

[Refer to Chapter S2A Discrete Random Variables Pg 11 for details.]

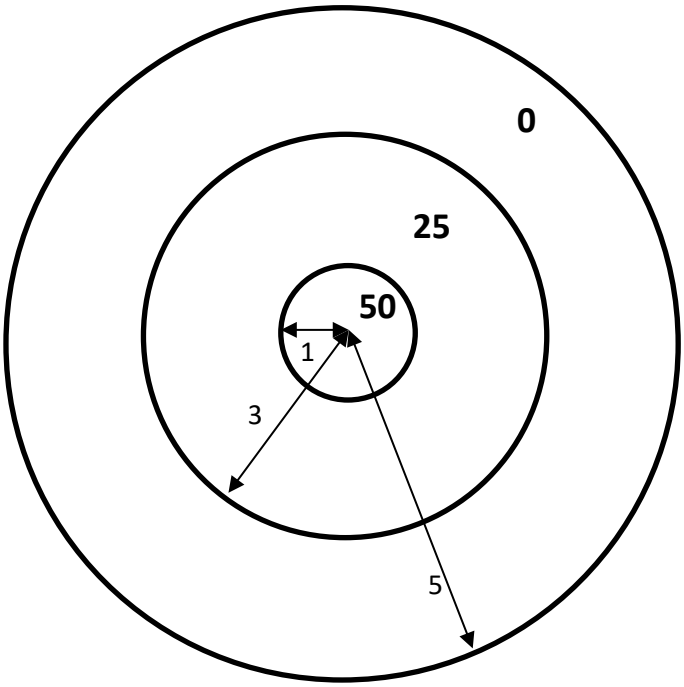
(iii)

$$\mathrm{Var}(Y) = \mathrm{Var}(2X_1 + X_2) = 2^2\mathrm{Var}(X_1) + \mathrm{Var}(X_2) = 2^2\mathrm{Var}(X) + \mathrm{Var}(X)$$

$$= 5\mathrm{Var}(X) = \frac{26}{5}$$

Source of Question: TMJC Prelim 9758/2023/02/Q6

3 At a particular game stall, players throw darts at a target board to earn points for redeeming prizes.



The target board can be modelled by three concentric circles with radii one unit, three units and five units respectively. A player will get a score of 50, 25 or 0 points if the dart lands on the respective regions labelled **50**, **25** or **0** as shown in the diagram above.

The probability that a dart lands on any region labelled **50**, **25** or **0** is proportional to the area of the region. It can be assumed that the borders between the regions have negligible thickness and the dart will only land on the target board, which has a total area of 25π units^2.

In a game, a player throws two darts at the target board, one after the other. The first dart will be removed before the next dart is thrown. Let S be the combined score that the player achieves from the two throws.

(a) Show that $\mathrm{P}(S=75)=\dfrac{16}{625}$. [3]

(b) Find the probability distribution of S. [4]

(c) Find the expectation and variance of S. [2]

Solution:

(a)

Area of target board $=\pi\left(5^2\right)=25\pi$

Area of region with score 50 $=\pi\left(1^2\right)=\pi$

Probability of dart hitting region with score 50 $=\dfrac{\pi}{25\pi}=\dfrac{1}{25}$

Area of region with score 25 $=\pi\left(3^2\right)-\pi=8\pi$

Probability of dart hitting region with score 25 $=\dfrac{8\pi}{25\pi}=\dfrac{8}{25}$

$$\mathrm{P}(S=75)=\frac{8}{25}\times\frac{1}{25}\times 2!$$
$$=\frac{16}{625} \quad (\text{shown})$$

(b)

Area of region with score 0 $=25\pi-9\pi=16\pi$

Probability of dart hitting region with score 0 $=\dfrac{16\pi}{25\pi}=\dfrac{16}{25}$

s	0	25	50	75	100
$\mathrm{P}(S=s)$	$\frac{16}{25}\times\frac{16}{25}$ $=\frac{256}{625}$	$\frac{16}{25}\times\frac{8}{25}\times 2!$ $=\frac{256}{625}$	$\frac{16}{25}\times\frac{1}{25}\times 2!$ $+\frac{8}{25}\times\frac{8}{25}$ $=\frac{96}{625}$	$\frac{16}{625}$	$\frac{1}{25}\times\frac{1}{25}$ $=\frac{1}{625}$

(c)

From GC,
$\mathrm{E}(S)=20$ and $\mathrm{Var}(S)=400$.

Source of Question: JPJC Prelim 9758/2023/02/Q10

4 The table below shows the BMI-for-age for boys aged 18 years in Singapore.

Weight Indicator / Age (years)	**Underweight**	**Overweight**
	$\leq$ 5th percentile	**$\geq$ 90th percentile**
18	≤ 16.7	$\geq m$

It is given that the BMI for 18-year-old boys follows a normal distribution $\mathrm{N}\left(22.7, \sigma^2\right)$.

(i) Show that $\sigma = 3.65$ correct to 2 decimal places. [2]

(ii) Hence, find the minimum value of BMI, m, such that a randomly chosen 18-year-old boy is overweight. [2]

The BMI for 15-year-old boys follows another normal distribution $\mathrm{N}\left(21.6, 3.49^2\right)$.

(iii) Find the probability that the average BMI of a randomly chosen 15-year-old boy and a randomly chosen 18-year-old boy is more than 20.1. [2]

(iv) State, in context, an assumption made in your calculation in **(iii)**. [1]

Solution:

(i)

Let X denote the BMI of **a** 18-year-old boy.

$$X \sim \mathrm{N}\left(22.7, \sigma^2\right)$$

$$\mathrm{P}\left(X \leq 16.7\right) = 0.05$$

$$\mathrm{P}\left(Z \leq \frac{16.7 - 22.7}{\sigma}\right) = 0.05$$

$$\frac{-6}{\sigma} \approx -1.6449$$

$$\sigma \approx 3.6476 = 3.65 \text{ (shown)}$$

(ii)

$$\mathrm{P}\left(X \geq m\right) = 0.1$$

Using GC, $m = 27.4$ (3 s.f)

A boy with BMI exceeding 27.4 will be at a higher percentile than the 90th percentile. Hence, the minimum value of BMI for a randomly chosen boy to be overweight is 27.4.

(iii)

Let Y denote the BMI of **a** 15-year-old boy.

$$Y \sim \mathrm{N}\left(21.6, 3.49^2\right)$$

Let $W = \dfrac{X + Y}{2}$

$$\mathrm{E}(W) = \frac{1}{2}\left[\mathrm{E}(X) + \mathrm{E}(Y)\right] = \frac{1}{2}\left(22.7 + 21.6\right) = 22.15$$

$$\mathrm{Var}(W) = \left(\frac{1}{2}\right)^2\left[\mathrm{Var}(X) + \mathrm{Var}(Y)\right] = \frac{1}{4}\left(3.6476^2 + 3.49^2\right) \approx 6.3713$$

	$W \sim \text{N}(22.15, 6.3713)$ $\text{P}(W > 20.1) \approx 0.792$
(iv)	The BMI of a randomly chosen 15-year-old boy and a randomly chosen 18-year-old boy are independent of each other.

Source of Question: RVHS Prelim 9758/2022/02/Q12 (modified)

5 The mass of a "Comforta" cushion has a normal distribution with mean 170g and standard deviation 12g.

(i) Find the probability that the mass of a randomly chosen "Comforta" cushion is more than 165g. [1]

(ii) Find the probability that, of three randomly chosen "Comforta" cushions, the mass of each cushion is less than 165g. [2]

Each "Comforta" cushion comes with a cushion cover. The mass of a cushion cover has a normal distribution with mean 20g and standard deviation σ.

(iii) It is given that more than 90% of "Comforta" cushions with their covers have mass at least 170g. Find the range of values of σ. [4]

(iv) Given that $\sigma = 8$. A "Comforta" cushion costs \$0.05 per gram to produce while a cushion cover costs \$0.02 per gram to produce. A "Comforta" cushion with cover is sold at \$10 each. Find the probability that profit will be made with the sale of a "Comforta" cushion with cover. [3]

The company manufacturing "Comforta" cushions decided to launch a new product called "Serena" cushions. The mean mass of a "Serena" cushion is 250g. A sample of 50 "Serena" cushions were collected for quality check and it was found that the sample standard deviation is 17g. Find the probability that the sample mean mass of a "Serena" cushion is at most 245g. [3]

Solution:

(i)	Let X be the mass of **a** "Comforta" cushion in grams. $X \sim \text{N}(170, 12^2)$ $\text{P}(X > 165) \approx 0.66154 = 0.662$ (3sf)
(ii)	$[\text{P}(X < 165)]^3$ $\approx (1-0.66154)^3$ $= 0.0388$ (3sf) <u>Alternative method</u> Let W be the **number** of cushions, out of 3, that has mass less than 165g. $W \sim \text{B}(3, 1-0.66154)$ $\text{P}(W = 3) = 0.0388$ (3sf)

(iii)

Let Y be the mass of **a** cushion cover in grams.

$Y \sim \mathrm{N}(20,\ \sigma^2)$

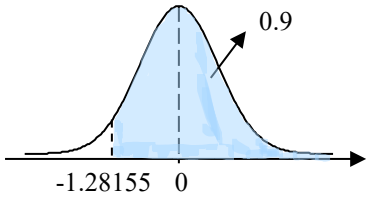
$X + Y \sim \mathrm{N}(190,\ 12^2 + \sigma^2)$

$\mathrm{P}(X + Y \geq 170) > 0.9$

$$\mathrm{P}\left(Z \geq \frac{170-190}{\sqrt{12^2+\sigma^2}}\right) > 0.9$$

From GC, $\mathrm{P}(Z \geq -1.28155) = 0.9$

From the diagram,



$$\frac{-20}{\sqrt{12^2+\sigma^2}} < -1.28155$$

$$\sqrt{12^2+\sigma^2} < \frac{20}{1.28155}$$

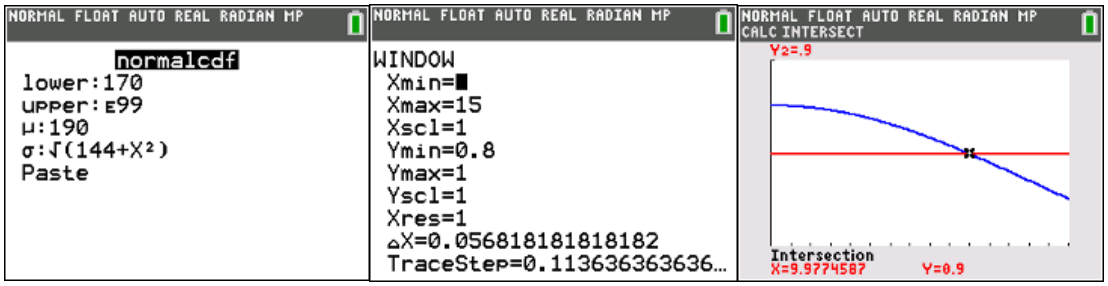
$$\sigma < 9.9775$$

$$\therefore 0 < \sigma < 9.98$$

Alternative method:

Key in Y1 = normalcdf (170, E99, 190, $\sqrt{12^2 + \mathrm{X}^2}$) where X is σ.

Find the intersection of Y1 with Y2 = 0.9. Remember to use appropriate window settings!



(iv)

Let W be the cost of **a** "Comforta" cushion with cover.

$W = 0.05X + 0.02Y$

$\mathrm{E}(W) = 0.05\mathrm{E}(X) + 0.02\mathrm{E}(Y) = 0.05(170)+0.02(20) = 8.9$

$\mathrm{Var}(W) = 0.05^2\mathrm{Var}(X) + 0.02^2\mathrm{Var}(Y) = 0.05^2(12^2)+0.02^2(8^2) = 0.3856$

i.e. $W \sim \mathrm{N}(8.9, 0.3856)$

$\mathrm{P}(W < 10) = 0.962$ (3sf)

Let S be the mass of **a** "Serena" cushion in grams.

Unbiased estimate of population variance, $s^2 = \frac{50}{49}\left(17^2\right) = \frac{14450}{49}$

Since $n = 50$ is large, by Central Limit Theorem, $\overline{S} \sim \mathrm{N}\left(250, \frac{\left(\frac{14450}{49}\right)^2}{50}\right)$ approximately.

$\mathrm{P}\left(\overline{S} \leq 245\right) = 0.0198$ (3sf)

Source of Question: YIJC Prelim 9758/2021/02/Q9

6 A chocolate manufacturer claims that the mean number of calories in a packet of chocolate is 300. A random sample of 50 packets of chocolate is selected and the number of calories, x, in each packet is measured. The results are summarised by:

$$\sum(x-300)=36, \quad \sum(x-300)^2=612.$$

(i) Find unbiased estimates of the population mean and variance. [2]

A nutritionist believes that the chocolate manufacturer has understated the mean number of calories in a packet of chocolate.

(ii) Test, at the 5% significance level, whether the nutritionist's belief is correct. [4]

(iii) Explain why there is no need for the nutritionist to know anything about the population distribution of the number of calories in the packets of chocolate. [1]

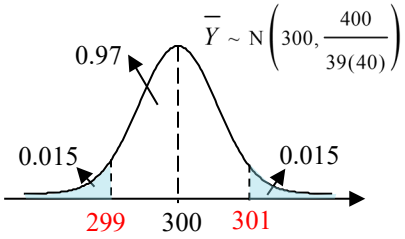
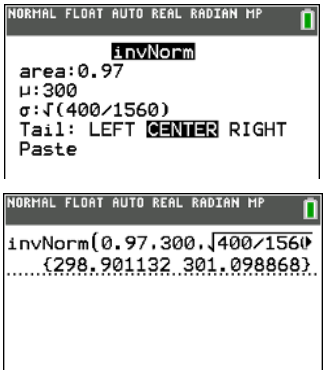
(iv) Explain, in the context of the question, the meaning of "at the 5% significance level". [1]

After receiving feedback from customers, the chocolate manufacturer improves the quality of the ingredients used. A random sample of 40 packets of the improved chocolate is taken. The mean number of calories of the 40 packets of chocolate is k and the variance is 10.

(v) Find the range of values of k such that there is sufficient evidence to conclude that the mean number of calories in a packet of the improved chocolate is not 300 at the 3% significance level. [4]

Solution:

(i)	Let $y=x-300$. Then $\overline{y}=\overline{x}-300$ and $s_x{}^2=s_y{}^2$. An unbiased estimate of population mean, μ, is $\overline{x}=\frac{36}{50}+300=300.72$ An unbiased estimate of population variance, σ^2, is $s^2=\frac{1}{49}\left(612-\frac{36^2}{50}\right)\approx 11.961=12.0(3\text{ s.f})$
(ii)	$H_0: \mu=300$ vs $H_1: \mu>300$ Perform a 1-tail test at 5% significance level. Under H_0, since $n=50$ is large, by Central Limit Theorem, $\overline{X}\sim N\left(300,\frac{11.961}{50}\right)$ approximately. Using a z-test, p-value $=P\left(\overline{X}>300.72\right)=0.0705$ (3 s.f.) Since the p-value = 0.0705 > 0.05, we **do not reject H_0** and conclude that there is **insufficient** evidence at the 5% level of significance that the nutritionist's belief is correct.

(iii)	Since the sample size of 50 is large, by Central Limit Theorem, the sample mean $\overline{X}$ follows a normal distribution approximately.
(iv)	There is a probability of 0.05 of wrongly concluding that the **mean** number of calories in a packet of chocolate is more than 300, when it is in fact 300.
(v)	Let Y be the number of calories in a packet of improved chocolate and μ_Y be the population mean of Y. Unbiased estimate of population variance, $s_Y^2 = \frac{40}{39}(10) = \frac{400}{39}$ $H_0: \mu_Y = 300$ vs $H_1: \mu_Y \neq 300$ Perform a 2-tail test at 3% level of significance. Under H_0, since $n = 40$ is large, by Central Limit Theorem, $\overline{Y} \sim N\left(300, \frac{400}{39(40)}\right)$ approximately. To **reject** H_0 at 3% level of significance, p-value ≤ 0.03 $P(\overline{Y} \leq k) \leq 0.015$ or $P(\overline{Y} \geq k) \leq 0.015$  From GC, $0 < k \leq 299$ or $k \geq 301$. GC screenshots: 

Source of Question: EJC Prelim 9758/2023/02/Q11

7 The amount customers spend is often correlated with the amount of time they spend in stores. According to a national survey of 500,000 shopping trips, the average time customers spend in a clothing store is 30 min. The first, second and third quartiles of the time spent, in min, are also given as follows:

Q_1	8.5
Q_2 (median)	18.5
Q_3	36.1

(a) Explain why the data suggests that shopping times do not follow a normal distribution.

[1]

Jack, a clothing store manager, suspects that customers spend less time in his store than the national average.

(b) Determine whether Jack should carry out a 1-tail or a 2-tail test. State the hypotheses for the test and define any symbols you use in this context. [3]

Jack collected a random sample of 50 shopping times. The results, x (in min), are summarised as follows:

$$\sum x = 1475, \quad \sum x^2 = 45029$$

(c) Test, at the 10% significance level, whether Jack's suspicion is valid. [5]

(d) Explain what is meant by 'at the 10% significance level' in the context of the question. [1]

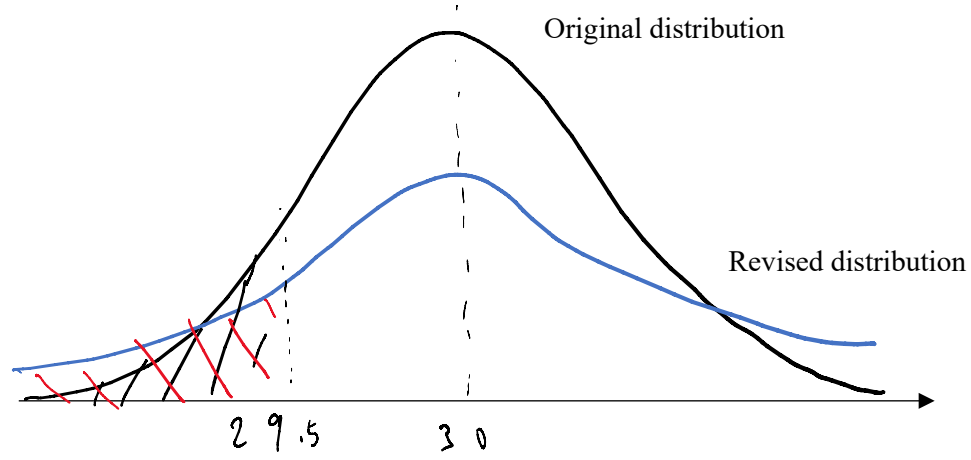
(e) Explain why in performing the hypothesis test, there is no need for Jack to be concerned that the shopping times of customers in his store do not follow a normal distribution. [1]

Upon closer inspection of the data, Jack realises that he has made some miscalculations. While there is no mistake in the calculation of $\sum x$, the value of $\sum x^2$ should be larger than initially calculated.

(f) Does this miscalculation affect Jack's initial conclusion? Explain your answer. [2]

Solution:

(a)	If shopping times follow a normal distribution, the median, $Q_2 = 18.5$, should be close to the mean of 30. But the mean is actually closer to Q_3 than Q_2. Alternatively Since a normal distribution is symmetrical, if shopping times follow a normal distribution, we should expect $Q_3 - Q_2$ and $Q_2 - Q_1$ to be roughly equal. But the latter is much larger than the former.
(b)	Jack should carry out a one-tail test since he is concerned whether customers spend less time in his store than the national average. $\mathrm{H}_0 : \mu = 30$ vs $\mathrm{H}_1 : \mu < 30$ where μ is the **population mean duration** customers <u>spend in his clothing shop</u> (in min).
(c)	$\overline{x} = \dfrac{1475}{50} = 29.5$ Unbiased estimate of population variance = $\dfrac{1}{49}\left(45029 - \dfrac{1475^2}{50}\right) = 30.949$ $\mathrm{H}_0 : \mu = 30$ vs $\mathrm{H}_1 : \mu < 30$ Perform 1-tail test at 10% level of significance.

	Under H_0, since $n = 50$ is large, by Central Limit Theorem $\overline{X} \sim N\left(30, \frac{30.949}{50}\right)$ approximately Using a z-test, p-value $= P\left(\overline{X} \leq 29.5\right) = 0.263$ (3 s.f) Since p-value > 0.10, we **do not reject** H_0 and conclude that there is **insufficient** evidence at 10% level of significance that the **mean** duration customers spend in his shop is less than 30 min.
(d)	It means that there is a probability of 0.10 of wrongly concluding that the **mean** shopping time in Jack's store is less than 30 min when it is in fact 30 min.
(e)	Jack does not need to be concerned that the distribution of shopping times in his shop does not follow a normal distribution as n = 50 is large and hence by Central Limit Theorem, the sample <u>mean</u> shopping time follows normally distribution approximately. (Thus the hypothesis test can still be carried out.)
(f)	Since $\sum x^2$ is larger, the revised value of s^2, i.e. the revised unbiased estimate of the population variance, will be larger. So the revised p-value – the proportion of the distribution that is less than 29.5 – will be larger than before. But the original p-value was already larger than 0.1, so the new p-value will continue to be larger than 0.1. //// --- Original p-value = 0.263 \\\\ --- New p-value > 0.263 (Compare the unshaded parts) 

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